Project: Waterfall

A generating station was built at the base of the falls to turn falling water into power for a continent. It stopped generating and it is still standing there. Project: Waterfall is the proposal to make it the home of Diaphora Labs.

Base of the falls · Niagara Falls, Ontario, Canada
Ontario Power Company generating station
Owned by the Niagara Parks Commission · out of service since 1999
Issued August 2026 · innovate@diaphoralabs.com

This is a proposal. Nothing described here is built, leased, bought or approved. The station is owned by the Niagara Parks Commission, and this document is an approach to them and to the public.

ONE

How the building got here

1890

The commission

Lord Kelvin's International Niagara Commission opened a worldwide call for proposals to harness the falls. The question was simple and enormous: how do you put this to work?

1896

Alternating current wins

Tesla and Westinghouse's polyphase alternating current was proven at Niagara. On 16 November 1896 power from the Adams plant reached Buffalo, twenty-two miles away, and transmitting power over distance stopped being a theory.

1905–1999

A working century

The Ontario Power Company station opened at the foot of the Horseshoe Falls in 1905. Fifteen generators produced 203,000 horsepower, converted from 25 to 60 hertz in the 1970s, and ran until the station was taken out of service in 1999.

TODAY

The building waits

The station still stands at the base of the falls, owned by the Niagara Parks Commission and out of service since 1999 — intact, dramatic, and unused, beside one of the most visited places on the continent.

PROPOSED

The second powering

Reopen it for the industries that will define the next hundred years — energy, water, computing, climate — with the public invited in. That is what this page is asking for.

TWO

One building, six uses

Every use below is proposed. None of it exists.

TURBINE HALL

The forum

Summits, exhibitions and public events under the original vaulted ceiling.

GENERATOR FLOORS

Labs and residencies

Bench and studio space for startups, university groups and corporate research teams.

PENSTOCK LEVELS	**The deep works** The shafts descending through the gorge become exhibition and testing space that cannot be built anywhere else.
SWITCH HOUSE	**Anchor studios** Signature space for a small number of institutions that want their name on the project.
GALLERY	**The record** The Tesla–Westinghouse story, told in the building where it happened, one floor from the work continuing it.
TERRACES	**The public edge** Cafés, events and open space facing the falls, so the campus stays open to the city that surrounds it.

THREE

What the site can test that nowhere else can

The conditions at the base of the falls — permanent mist, natural spray icing, a constant acoustic floor, a sustained grade and the station's own below-grade works — are a set of test environments that cannot be manufactured. Each is subject to survey and to Niagara Parks' consent.

MIST AND ATMOSPHERE	**The fog corridor** Lidar and camera validation in natural, repeatable fog; accelerated weathering racks in the spray line.
ICE AND WINTER	**The icing station** De-icing systems and icephobic coatings under natural spray icing; heat pumps certified in a real winter.
SOUND AND VIBRATION	**The noise floor** Noise-cancelling audio and hearing technology proven against a constant roar; acoustic models trained on the steadiest noise on earth.
UNDERGROUND	**The tunnel range** GPS-denied robotics and confined-space rescue training in the station's below-grade works, subject to survey.
WATER AND POWER	**The hydraulic loop** Turbine and fish-passage certification in the original penstocks; a hydro-powered hydrogen demonstration.

PEOPLE, GRADE AND CABLE

The living lab
Opt-in crowd-flow and accessibility research at world scale; EV braking trials on a sustained grade.

FOUR

Niagara-Proven

A proposed certification mark. A product that has been through the ranges above carries Niagara-Proven — evidence it was tested against real conditions rather than simulated ones. The mark does not exist yet and nothing has been certified.

FIVE

Who supports Diaphora Labs

These organisations have backed the founders and support Diaphora Labs. None of them is an investor or funder, none has committed capital or space, and none has committed anything to Project: Waterfall.

Niagara Falls Innovation Hub
Brock University
Velocity

SIX

Where these facts come from

THE STATION: 1905 TO 1999, 203,000 HP
Ontario Power Company Generating Station
https://en.wikipedia.org/wiki/Ontario_Power_Company_Generating_Station

OWNED BY NIAGARA PARKS, OUT OF SERVICE
Niagara Parks, Interests in Redevelopment of Inactive Power Stations
https://www.niagaraparks.com/media-room/news/power-stations-redevelopment

THE 1890 COMMISSION
IEEE, Milestones: Adams Hydroelectric Generating Plant
https://ethw.org/Milestones:Adams_Hydroelectric_Generating_Plant,_1895

POWER REACHING BUFFALO, 1896
Engineering and Technology History Wiki, Early Electrification of Buffalo
https://ethw.org/Early_Electrification_of_Buffalo

SEVEN

What we are asking for

A conversation with the Niagara Parks Commission about the future of the Ontario Power Company station, and public support for reopening it as a working institution rather than a monument that costs money to stand still.

If you are a partner, a design or engineering firm, a startup or researcher, a Niagara local, an investor or anchor tenant, or a journalist, there is a way to sign on at diaphoralabs.com/waterfall.